

VATSAL AGRAWAL

AI/ML Engineer · Full-Stack Developer · Co-Founder, AISkillBench
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SUMMARY

Computer Science (AI & ML) undergraduate at VIT Chennai and co-founder of AISkillBench. I build and ship production AI systems end-to-end — including a transformer language model written from scratch in PyTorch, multi-agent LLM platforms, and full-stack SaaS — several deployed and live.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Dart, C/C++, Java, SQL / PL-SQL, HTML, CSS

AI / ML: Transformers & LLMs (built from scratch in PyTorch), Multi-Agent Systems (LangGraph, LangChain), RAG & Vector Stores (ChromaDB), Computer Vision (OpenCV), OCR, Prompt Engineering, LLM Guardrails, NEAT

Frameworks & Libraries: React, Next.js, FastAPI, Flask, Flutter (Riverpod), Node.js, PyTorch, NumPy, Playwright, Pygame, Streamlit

Backend & Infrastructure: PostgreSQL, Supabase (Auth / Realtime / Edge Functions), WebRTC, REST APIs, nginx, systemd, GitHub Actions, Netlify, Railway

Tools & Platforms: Git / GitHub, Claude Code, Windsurf, Apify, Buffer, n8n

Concepts: Data Structures & Algorithms, OOP, System Design

EDUCATION

B.Tech, Computer Science & Engineering (AI & Machine Learning) 2024 – 2028

Vellore Institute of Technology (VIT), Chennai · CGPA: 8.56 / 10

Relevant coursework: Artificial Intelligence, DBMS, Theory of Computation, Computer Networks, Linear Algebra & Complex Analysis.

EXPERIENCE

Co-Founder & Full-Stack / AI Engineer, AISkillBench, Nov 2025 – Present

AI Skills Learning & Certification Platform (live) · React, TypeScript, Supabase, Python, FastAPI

- Co-founded and built an end-to-end live EdTech platform that generates personalised AI-skill learning paths (Fields → Roles → Skills → Courses) via a goal-weighted scoring algorithm, with QR-verified certifications.
- Shipped three production AI agents: a learning coach grounded in each user's live path, a RAG support bot (TF-IDF + ChromaDB with deterministic payment/progress lookups), and a multi-LLM email agent with classifier and reviewer sub-agents.
- Built a 5-layer LLM guardrail framework with a regression suite, and integrated multi-currency payments (Razorpay, PhonePe, Stripe).
- Deployed on an Ubuntu VPS (nginx, systemd) across separate dev and production Supabase environments.

AI / Software Engineering Intern, Atelia.ai, Jun 2026 – Present

London, United Kingdom (Remote) · Flutter, Riverpod, Material 3, Dart

- Build and ship enhancements to the company's consumer-facing (B2C) Flutter mobile app, including a multi-shot garment-scan flow that returns a fabric-quality score and matches products with deep links, plus a guided in-scan chat that prompts users to confirm or correct detected details before the final result.
- Implemented a resilient product-recommendation flow with layered fallbacks and HEAD-validated URL resolution across a 24-retailer brand table, built on a custom design system implementing the company's brand book.
- Contribute beyond engineering in a founding-stage team: feature testing and bug triage/resolution tracking, demo-ready builds, product and feature documentation, competitor and market research, and growth content for LinkedIn and Instagram.

Analytics & Growth Collaborator, Dietichen (YouTube Channel) 2022 – Present

- Analysed audience retention, algorithmic signals, and publishing patterns to shape data-driven content strategies, supporting steady subscriber and viewership growth.

PROJECTS

Verified Code-Optimisation Engine: From-Scratch SLM 2026

Python, PyTorch · github.com/vatsal-agra/SLM-code-optimizer

- Built a 33.9M-parameter decoder-only transformer from scratch in PyTorch (no HuggingFace or pretrained weights) — custom BPE tokeniser, RoPE, multi-head causal attention, RMSNorm, SwiGLU — trained end-to-end on a single 4GB laptop GPU with fp16, gradient accumulation, and cosine LR scheduling.
- Layered a model-agnostic verification engine that proves each rewrite is behaviorally identical to the original and measurably faster, validating candidates against hundreds of fuzzed inputs in isolated subprocesses and fitting complexity curves to timing data before accepting.
- Productized into an AST-based hotspot scanner with pluggable proposers (local SLM, Claude API, or rule-based), repo-wide scanning, safe auto-fix, HTML evidence reports, and GitHub Actions CI.

AUTOTING: Autonomous AI Marketing&SEO Platform 2026

Next.js, FastAPI, Supabase, Gemini

- Full-stack marketing SaaS with a brand-strategy layer most AI tools lack: it crawls a company's site to infer a structured "Brand Soul" (belief, audience, voice, visual identity) via Simon Sinek's Golden Circle, then generates on-brand content led by the brand's WHY.
- Engineered an agentic content pipeline (brand-archaeology → strategist → copywriter → film-director) with a two-stage image engine and a vision-judge scoring best-of-N outputs.
- Shipped a 9-tool SEO suite and multi-channel publishing (Buffer, X, Instagram, Medium), with SSRF-guarded crawling and free-tier LLM key rotation.

Autonomous AI Newsletter Platform 2026 · Live

Next.js, FastAPI, LangGraph, Supabase

- Built a LangGraph multi-agent pipeline (Scout → Anchor → Copy Editor → Publisher) that researches, writes, adversarially reviews, and publishes weekly newsletters autonomously; the Copy Editor loops drafts back until they pass quality thresholds.
- Shipped with a Next.js subscriber site, multi-LLM fallback routing, ChromaDB cross-issue memory to avoid repetition, AI hero images, and category-filtered email via Resend with an admin panel for scheduling and spend tracking.

Dice Alley: Real-Time Multiplayer Game Platform 2025 · Live

Next.js, TypeScript, Supabase, WebRTC

- Built a browser-based game-night platform with 9 multiplayer games, each an independent TypeScript state machine synced via a host-authoritative engine over Supabase Realtime (move sync, late-join resync, reconnect-from-snapshot).
- Built peer-to-peer voice chat from scratch (WebRTC full mesh, STUN/TURN) plus XP/levelling, 30+ achievements, a live leaderboard, and guest play with Postgres row-level security.

Jarvis: AI Voice Assistant & Autonomous Agent (Windows) 2025

Python, Gemini, Playwright

- Built a ~5,300-line desktop agent exposing ~100 callable tools across voice, vision, web automation, desktop control, and memory, running on free-tier LLMs via OpenAI-compatible function calling with multi-key/multi-model rotation.
- Drives any website via an indexed-DOM snapshot with vision-based verification, controls native Windows apps via UI-Automation, and adds long-term RAG memory, runtime self-extension, Telegram phone control, and a live SSE HUD.

ACHIEVEMENTS & HIGHLIGHTS

- 2nd Place, GDG FlutterSprint Hackathon (built MockMate), organised by Google Developers Group.
- Solo builder shipping across AI/ML, full-stack web, mobile, and automation, with several projects deployed to production.